

FloodSense User Survey

Comprehensive Analysis Report

Bacoor City Stakeholder Survey | n = 40 Respondents

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Specialization: Geospatial Analysis | DRRM | Mobile UX/UI

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CATEGORICAL SURVEY ANALYSIS

SECTION A: Respondent Profile (Q1 to Q3)

Q1: Which barangay in Bacoor City do you currently reside in?

Data Summary:

Barangay	Count	%
Soldiers 4	10	25.0%
Talaba 1	5	12.5%
Molino 3	5	12.5%
Niog 1	4	10.0%
Zapote 1	3	7.5%
Molino 2	2	5.0%
Niog 2	2	5.0%
Molino 4	2	5.0%
Mambog 2	2	5.0%
Mambog	1	2.5%
Magdiwang	1	2.5%
Soldiers Hills	1	2.5%
Habay 1	1	2.5%
Mambog 1	1	2.5%
Total	40	100%

Key Finding:

The respondent pool is geographically diverse, spanning 14 barangays across Bacoor City, though heavily weighted toward Soldiers 4 (25%), which alone accounts for one quarter of responses. The Molino cluster (Molino 2, 3, 4) collectively represents 22.5% of respondents, and the Talaba-Niog corridor contributes another 27.5%. This geographic spread is analytically valuable — it samples multiple flood-risk zones across the city rather than a single locality — and confirms that the survey captures a cross-barangay perspective. However, the disproportionate representation of Soldiers 4 means its residents' experiences carry greater statistical weight in aggregate findings.

Q2: How often does your area experience flooding?

Data Summary:

Frequency	Count	%
Almost every rainy season	31	77.5%
Occasionally (once every few years)	4	10.0%
Rarely	4	10.0%
Never	1	2.5%
Total	40	100%

Key Finding:

An overwhelming 77.5% of respondents report that their areas flood "almost every rainy season," establishing that the survey population is predominantly high-frequency flood-affected residents. Only 1 respondent (2.5%) reported never experiencing flooding. This is not a community speculating about an abstract hazard — these are people living in recurrent, seasonal flood exposure. This level of chronic flood exposure renders the community acutely vulnerable and underscores an urgent, non-hypothetical need for flood information tools.

Q3: How would you describe your current level of awareness about which areas in Bacoar City are flood-prone?

Data Summary:

Awareness Level	Count	%
Somewhat aware	16	40.0%
Not sure	10	25.0%
Well-informed	9	22.5%
Not aware at all	5	12.5%
Total	40	100%

Key Finding:

Only 22.5% of respondents consider themselves "well-informed" about flood-prone areas in the city. A combined 77.5% range from "somewhat aware" to "not aware at all," with 25% in an explicitly uncertain state of "not sure." This paradox — intimate familiarity with flooding as a recurring experience, yet lacking spatial literacy to identify which areas are at risk — directly validates the core premise of FloodSense.

SECTION B: Current Flood Information Experience (Q4 to Q5)

Q4: How satisfied are you with the flood-related information currently available to you?

Data Summary (1 = Very Dissatisfied, 4 = Very Satisfied):

Rating	Count	%
1 – Very Dissatisfied	8	20.0%
2 – Dissatisfied	18	45.0%
3 – Somewhat Satisfied	9	22.5%
4 – Very Satisfied	5	12.5%
Total	40	100%
Mean	2.28 / 4	—

Key Finding:

65.0% of respondents (26 of 40) rate their satisfaction with existing flood information sources as low (ratings 1 or 2), with 45.0% selecting "Dissatisfied." The mean score of 2.28 out of 4 falls below the midpoint, indicating community-wide structural dissatisfaction with current channels. This widespread dissatisfaction is the foundational demand signal that justifies a purpose-built flood information platform for Bacoar City.

Q5: Have you ever made a wrong or unsafe decision during a flood event due to a lack of accurate information?

Data Summary:

Response	Count	%
Yes	29	72.5%
I am not sure	9	22.5%
No	2	5.0%
Total	40	100%

Key Finding:

72.5% of respondents confirm they have made a wrong or unsafe decision during a flood event as a direct result of information gaps. An additional 22.5% were "not sure," suggesting the actual figure may approach or exceed 90%. Only 2 respondents (5.0%) definitively reported making no wrong decision. Information failure in this community is not merely inconvenient — it is physically dangerous and has already resulted in known unsafe behaviors for nearly three-quarters of the sample population.

SECTION C: Flood Risk Awareness (Q6 to Q8)

Q6: How difficult is it to find which specific areas in Bacoar City are at risk under different rainfall conditions?

Data Summary (1 = Very Easy, 5 = Very Difficult):

Rating	Count	%
1 – Very Easy	3	7.5%
2 – Easy	12	30.0%
3 – Moderate	11	27.5%
4 – Difficult	11	27.5%
5 – Very Difficult	3	7.5%
Total	40	100%
Mean	2.98 / 5	—

Key Finding:

The mean of 2.98 positions this question nearly at the midpoint of the difficulty scale, reflecting a deeply divided community: 35.0% find accessing barangay-specific flood risk information difficult or very difficult, while 37.5% find it easy or very easy. This polarization suggests that information accessibility in Bacoor City is inconsistent — likely tied to barangay-level variation in official communication. A standardized, app-based platform would eliminate this informational inequality.

Q7: How confident are you in knowing what PAGASA reports mean for flood risk in your specific barangay?

Data Summary (1 = Not at All Confident, 5 = Extremely Confident):

Rating	Count	%
1 – Not at All Confident	8	20.0%
2 – Slightly Confident	11	27.5%
3 – Somewhat Confident	14	35.0%
4 – Mostly Confident	6	15.0%
5 – Extremely Confident	1	2.5%
Total	40	100%
Mean	2.52 / 5	—

Key Finding:

47.5% of respondents (19 of 40) have low confidence (ratings 1 or 2) in translating PAGASA regional weather advisories into actionable, barangay-specific flood risk understanding. Only 17.5% express high confidence (ratings 4 or 5). The mean of 2.52 signals a significant translation gap — this is precisely the gap that FloodSense's ML-based susceptibility mapping is designed to bridge.

Q8: How useful would it be to visually see which barangays in Bacoor City are at low, moderate, high, or very high flood risk — all in one place?

Data Summary (1 = Not at All Useful, 5 = Extremely Useful):

Rating	Count	%
1 – Not at All Useful	1	2.5%
2 – Slightly Useful	6	15.0%
3 – Somewhat Useful	7	17.5%
4 – Mostly Useful	7	17.5%
5 – Extremely Useful	19	47.5%
Total	40	100%
Mean	3.93 / 5	—

Key Finding:

65.0% of respondents (26 of 40) rate this feature as highly useful (ratings 4 or 5), with nearly half the entire sample (47.5%) selecting "Extremely Useful" — the single most concentrated response across all quantitative questions. The mean of 3.93 is the second-highest utility score in the survey. This is strong market validation for a unified, visual flood risk dashboard.

SECTION D: Personal Safety & Location Awareness (Q9 to Q11)

Q9: How difficult is it to find out whether your specific barangay is at risk during a flood event or heavy rainfall?

Data Summary (1 = Very Easy, 5 = Very Difficult):

Rating	Count	%
1 – Very Easy	2	5.0%
2 – Easy	6	15.0%
3 – Moderate	9	22.5%
4 – Difficult	13	32.5%
5 – Very Difficult	10	25.0%
Total	40	100%
Mean	3.58 / 5	—

Key Finding:

57.5% of respondents (23 of 40) report difficulty (ratings 4 or 5) in determining whether their own specific barangay is at risk during an active flood or rainfall event. The mean of 3.58 tilts decisively toward the difficult end. The inability to answer the simple question "Is my barangay at risk right now?" in real time is a foundational gap that directly endangers lives.

Q10: When flood risk information is available, how clear and easy to understand is it for you?

Data Summary (1 = Very Unclear, 5 = Very Clear):

Rating	Count	%
1 – Very Unclear	6	15.0%
2 – Somewhat Unclear	10	25.0%
3 – Neutral	10	25.0%
4 – Somewhat Clear	8	20.0%
5 – Very Clear	6	15.0%
Total	40	100%
Mean	2.95 / 5	—

Key Finding:

40.0% of respondents (16 of 40) find available flood risk information unclear or very unclear (ratings 1 or 2). An additional 25% are neutral — meaning only 35.0% find information positively clear. The clarity gap amplifies the availability gap: not only is information hard to find, but when it arrives, it often fails to communicate actionably. This finding directly justifies FloodSense's plain-language, ML-translated risk messaging feature.

Q11: How helpful would it be to receive a plain-language message telling you the exact flood risk level and what to expect under a specific rainfall condition?**Data Summary (1 = Not at All Helpful, 5 = Extremely Helpful):**

Rating	Count	%
2 – Slightly Helpful	3	7.5%
3 – Somewhat Helpful	15	37.5%
4 – Mostly Helpful	5	12.5%
5 – Extremely Helpful	17	42.5%
Total	40	100%
Mean	3.90 / 5	—

Note: No respondents selected Rating 1 for this item.

Key Finding:

55.0% of respondents (22 of 40) rate this feature as highly helpful (ratings 4 or 5), with 42.5% selecting "Extremely Helpful." Crucially, zero respondents selected "Not at All Helpful" — a unanimous signal that even the most skeptical respondents see some value in plain-language risk translation. The mean of 3.90 is among the highest utility scores in the survey.

SECTION E: Preparedness & Knowing What to Do (Q12 to Q13)

Q12: When flooding is expected, how confident are you in knowing exactly what steps to take to protect yourself and your family?

Data Summary (1 = Not at All Confident, 5 = Extremely Confident):

Rating	Count	%
1 – Not at All Confident	7	17.5%
2 – Slightly Confident	9	22.5%
3 – Somewhat Confident	16	40.0%
4 – Mostly Confident	5	12.5%
5 – Extremely Confident	3	7.5%
Total	40	100%
Mean	2.70 / 5	—

Key Finding:

40.0% of respondents (16 of 40) report low confidence (ratings 1 or 2) in knowing what protective steps to take when flooding is expected. Only 20.0% express high confidence (ratings 4 or 5). The dominant response is "Somewhat Confident" (40.0%), which — in a genuine emergency — represents a dangerous gap. A community that is only partially sure of its protective protocols is one that hesitates, and hesitation during a rapid-onset flood event can be fatal.

Q13: How helpful would it be to have a clear, step-by-step list of actions based on how severe the flood risk is in your area?

Data Summary (1 = Not at All Helpful, 5 = Extremely Helpful):

Rating	Count	%
1 – Not at All Helpful	1	2.5%
2 – Slightly Helpful	3	7.5%
3 – Somewhat Helpful	6	15.0%
4 – Mostly Helpful	7	17.5%
5 – Extremely Helpful	23	57.5%
Total	40	100%
Mean	4.20 / 5	—

Key Finding:

With a mean of 4.20 — the highest utility score in the entire survey — and 75.0% of respondents (30 of 40) selecting ratings 4 or 5, this feature concept achieved near-consensus endorsement. The single most desired FloodSense feature, by statistical weight, is a tiered, risk-severity-calibrated

action checklist. The demand is for prescribed, escalating action protocols that remove guesswork from the emergency response chain.

SECTION F: Evacuation (Q14 to Q15)

Q14: How difficult is it to find out where the nearest evacuation center is from your current location during a flood threat?

Data Summary (1 = Very Easy, 5 = Very Difficult):

Rating	Count	%
1 – Very Easy	3	7.5%
2 – Easy	6	15.0%
3 – Moderate	10	25.0%
4 – Difficult	10	25.0%
5 – Very Difficult	11	27.5%
Total	40	100%
Mean	3.50 / 5	—

Key Finding:

52.5% of respondents (21 of 40) find locating the nearest BDRRMO-designated evacuation center difficult or very difficult during an active flood threat. Only 22.5% find it easy or very easy. Evacuation center inaccessibility is not an abstract planning problem — it is a real-time navigational failure that directly prevents people from reaching safety.

Q15: How helpful would it be to automatically know the nearest BDRRMO-designated evacuation centers ranked by distance, with directions?

Data Summary (1 = Not at All Helpful, 5 = Extremely Helpful):

Rating	Count	%
1 – Not at All Helpful	1	2.5%
2 – Slightly Helpful	1	2.5%
3 – Somewhat Helpful	8	20.0%
4 – Mostly Helpful	11	27.5%
5 – Extremely Helpful	19	47.5%
Total	40	100%
Mean	4.15 / 5	—

Key Finding:

75.0% of respondents (30 of 40) rate proximity-ranked evacuation center navigation as highly helpful (ratings 4 or 5), with a mean of 4.15 and 47.5% selecting "Extremely Helpful." Combined with Q14's difficulty data, this forms a clear demand-supply gap: residents urgently need evacuation center proximity information, cannot currently access it, and will strongly value an app that provides it automatically.

SECTION G: Sharing Flood Information (Q16 to Q17)

Q16: How often do you share flood-related warnings or updates with family members or neighbors when you receive them?

Data Summary:

Frequency	Count	%
Most of the time	18	45.0%
Sometimes	13	32.5%
Always	5	12.5%
Rarely	4	10.0%
Total	40	100%

Key Finding:

57.5% of respondents (23 of 40) share flood warnings "Most of the time" or "Always," and an additional 32.5% share "Sometimes." Only 10.0% rarely share warnings. Bacoor City residents are instinctively prosocial information relayers. A sharing feature built into FloodSense will not need to teach new behavior — it will digitize and amplify an existing social practice.

Q17: How helpful would it be to quickly send a flood risk report for any barangay to a family member or neighbor through Viber, WhatsApp, or SMS?

Data Summary (1 = Not at All Helpful, 5 = Extremely Helpful):

Rating	Count	%
1 – Not at All Helpful	1	2.5%
2 – Slightly Helpful	3	7.5%
3 – Somewhat Helpful	10	25.0%
4 – Mostly Helpful	7	17.5%
5 – Extremely Helpful	19	47.5%
Total	40	100%
Mean	4.00 / 5	—

Key Finding:

65.0% of respondents (26 of 40) rate one-tap sharing of barangay flood risk reports as highly helpful (ratings 4 or 5), with a mean of 4.00. Combined with Q16's finding that most residents already share warnings actively, this represents a compound demand: residents who already share would do so more frequently, more accurately, and more usefully if given a purpose-built, one-tap sharing mechanism.

SECTION H: Open-Ended Feedback (Q18 to Q19)

1. Thematic Coding Matrix

THEME 1: Real-Time, Hyperlocal Flood Status (Dominant Theme)

Frequency: ~22 mentions across Q18 & Q19

Residents across multiple barangays describe the same core frustration: flood information, when it exists, is too broad, too delayed, or too general to be actionable at the street level. They need to know which specific roads are already flooded, how deep the water is, and whether passage remains safe — in real time.

Representative responses:

"Real-time updates on which specific streets are already flooded and how deep the water is. (Q18)"

"Live, street-level flood maps showing exactly which specific roads are still passable in real time. (Q19)"

THEME 2: Advance Warning Failure — No Predictive Foresight

Frequency: ~18 mentions across Q18 & Q19

Multiple respondents express that they are never informed in advance about when or where flooding will occur. They describe situations where they had no warning before water rose rapidly around their homes.

Representative responses:

"Hindi kami nabibigyan ng malinaw na babala bago dumating ang baha — walang eksaktong oras o lugar na ibinibigay. (Q18)"

"By informing before the heavy rain. (Q19)"

THEME 3: Evacuation Center & Emergency Hotline Inaccessibility

Frequency: ~14 mentions across Q18 & Q19

Residents report that during active flood events, they cannot find the locations of operational evacuation centers or the contact numbers of local emergency response units. Several respondents explicitly call for a single, consolidated resource containing evacuation center locations, directions, and hotlines.

Representative responses:

"During the flood, wala kaming mahanap na listahan ng evacuation centers o hotlines na malapit sa amin. (Q18)"

"Finding evacuation area. (Q18)"

THEME 4: Mismatch Between Official Signals and Ground Reality

Frequency: ~5 mentions across Q18 & Q19

A notable subset of respondents articulate a specific credibility gap: official weather signals and news reports do not match what they actually experience on the ground. This disconnect erodes institutional trust and leads residents to discount official warnings.

Representative responses:

"The news says there's a light rain but in reality it's the opposite. (Q18)"

"The news announces that the storm is only under signal no. 1, but what I experience feels more like signal no. 3 or 4 because the flooding is too high. (Q18)"

THEME 5: Demand for a Unified, Consolidated Flood Information Platform

Frequency: ~12 mentions in Q19

Across both English and Filipino responses, residents consistently converge on the same solution concept: a single, authoritative, always-current resource where all flood-related information is available in one place.

Representative responses:

"A dedicated text blast system or mobile app that sends immediate water level alerts for our specific barangay. (Q19)"

"Makakatulong talaga kung may isang sistema o app na nagbibigay ng kumpletong flood-related na impormasyon. (Q19)"

2. Solid Narrative Summary

The open-ended responses from Bacoar City residents paint a consistent and urgent portrait of a community that has normalized living with flood risk — yet remains critically under-informed at every stage of the flood cycle. The dominant emotional register across both Filipino and English responses is one of resigned familiarity combined with barely suppressed anxiety.

The most pervasive and deeply felt gap is the absence of hyperlocal, real-time information. Residents do not need to be told that "heavy rainfall is expected" — they can feel it. What they desperately lack is the translation of that meteorological reality into ground-truth street-level data: which roads are impassable, how quickly the water is rising, and which direction safety lies.

A striking pattern emerges in the Tagalog-language responses: a culture of sudden onset. Respondents describe scenarios where water rises around them before any warning reaches them. The phrase "bigla na lang" (suddenly/without warning) appears repeatedly, suggesting that rapid-onset flooding without advance notification is not an exceptional event but a routine one. This reveals a structural latency problem in the existing warning chain.

The second major theme — evacuation center inaccessibility — reveals that even when residents decide to evacuate, they face a navigational information vacuum. They do not know where the nearest operational center is, whether it is accessible given current flood conditions, or who to call for guidance.

Finally, the institutional trust deficit exposed in Theme 4 has significant implications for any FloodSense deployment strategy. Residents who have repeatedly found official warnings inconsistent with ground reality will be predisposed to distrust new information systems unless they demonstrably outperform existing channels.

3. Critical Qualitative Findings

Gap 1 — The Latency Crisis: The word "real-time" appears in multiple independent responses, confirming that delayed information is operationally useless. By the time announcements reach residents, conditions have already changed. Any viable solution must deliver information in sub-minute intervals during active events.

Gap 2 — The Specificity Vacuum: Existing systems communicate at the city or region level; residents need block-by-block intelligence. The scale mismatch between available information and decision-making needs is a fundamental design failure in current systems.

Gap 3 — The Evacuation Navigation Void: "Wala kaming mahanap na tamang hotline o evacuation site." During the worst moments of a flood, residents are left without a lifeline to reach safety. The absence of a single, authoritative, real-time evacuation directory is not a gap — it is a crisis.

Gap 4 — The Credibility Collapse: When the gap between official communication and lived experience is large, residents begin to discount all official warnings — including legitimate ones. Rebuilding credibility requires a system that consistently validates ground truth.

Gap 5 — Fatalistic Adaptation: "Wala. Sanay na mga tao dito na bahain e." [Nothing. The people here are used to flooding.] This single response encapsulates a dangerous cognitive adaptation. Any FloodSense deployment must actively re-engage these fatalistic users by demonstrating tangible, early protective value.

PART 2: PROPOSED SOLUTION CONCEPT & INTEGRATION

Working Title: FloodSense — An Augmented Reality Mobile Application with Machine Learning-Based Flood Susceptibility Mapping for Bacoor City

Pillar 1: Data-Driven Justification

The FloodSense concept is not a solution searching for a problem — it is a direct, evidence-grounded response to a documented, quantified crisis in Bacoor City's flood information ecosystem.

- The Scale of Exposure is Non-Negotiable. Q2 establishes that 77.5% of surveyed residents experience flooding almost every rainy season. FloodSense must be designed as a year-round tool that residents habituate to — not a one-time crisis app.
- The Information Infrastructure Has Already Failed, Repeatedly. Q5 reveals that 72.5% of respondents have made dangerous decisions due to information failure. The urgency of FloodSense is not anticipatory; it is remedial.
- Dissatisfaction with Existing Channels is Overwhelming. Q4's 65.0% dissatisfaction rate across all current sources signals that incremental improvements are insufficient. A fundamentally new platform is warranted.
- Spatial Knowledge Gaps Are Dangerous. Q3 shows that 77.5% of residents cannot confidently identify which areas are flood-prone. Q9 confirms that 57.5% cannot determine their own barangay's risk status during an active event.
- PAGASA Advisories are Being Lost in Translation. Q7's finding that 47.5% cannot translate national weather warnings into local flood risk meaning confirms that the problem is not only information availability but interpretation.
- Evacuation Infrastructure is Functionally Inaccessible. Q14's 52.5% difficulty rate reveals that the BDRMO's physical infrastructure exists but is informationally invisible to half the population during the critical window when it is needed most.
- The Community Demand Signal is Extraordinary. Q13 (mean 4.20), Q15 (mean 4.15), Q17 (mean 4.00), Q11 (mean 3.90), and Q8 (mean 3.93) all exceed 3.90 on 5-point utility scales — strong, consistent expressions of urgent need.

Pillar 2: Feature-to-Data Mapping

Survey Finding	Data Point	Recommended FloodSense Feature
Q9: 57.5% cannot determine their barangay's risk during active events	Mean 3.58 difficulty	Real-time AR Flood Risk Overlay — Camera feed overlaid with live, color-coded risk level for the user's GPS-pinpointed barangay

Survey Finding	Data Point	Recommended FloodSense Feature
Q8: 65.0% strongly want unified visual flood risk dashboard	Mean 3.93; 47.5% = Extremely Useful	ML-Powered Flood Susceptibility Map — City-wide, color-coded barangay risk map (Low/Moderate/High/Very High)
Q7: 47.5% cannot translate PAGASA advisories into local risk	Mean 2.52 confidence	PAGASA-to-FloodSense Signal Translator — ML model converting raw PAGASA rainfall forecasts into barangay-specific, plain-language levels
Q10: 40.0% find available information unclear	Mean 2.95 clarity	Plain-Language Risk Bulletin Generator — Auto-generated, jargon-free push notification with specific risk level and expected conditions
Q11: 55.0% rate plain-language risk messages as highly helpful	Mean 3.90 utility	Location-Aware Push Notification System — GPS-triggered alerts specific to the user's current barangay
Q13: 75.0% want tiered action checklists (highest mean: 4.20)	57.5% = Extremely Helpful	Tiered DRRM Action Protocol Module — Risk-severity-indexed action guides from LOW to VERY HIGH risk
Q14: 52.5% cannot find nearest evacuation center during emergencies	Mean 3.50 difficulty	Proximity-Ranked Evacuation Center Navigator — GPS-based auto-ranking with walking/driving routes
Q15: 75.0% rate automated evacuation routing as highly helpful	Mean 4.15 utility	AR Evacuation Pathfinding — Camera overlay displaying directional arrows to nearest open evacuation center
Q16: 57.5% already share warnings most of the time or always	Active prosocial sharing behavior	One-Tap Risk Report Sharing Module — Pre-formatted barangay risk snapshot shareable to Viber, WhatsApp, or SMS
Q17: 65.0% want messaging-app integration for sharing	Mean 4.00 utility	Deep-Link Sharing API Integration — Native Viber/WhatsApp/SMS integration with auto-generated flood risk messages
Q5: 72.5% made unsafe decisions due to information gaps	Documented harm event	Hazardous Route Warning System — Real-time alert when user attempts navigation through a high-risk or flooded barangay

Survey Finding	Data Point	Recommended FloodSense Feature
Q18/Q19 Themes: Demand for hyperlocal, real-time street flood status	Recurrent qualitative demand	Community Flood Report Feature — Crowdsourced, GPS-tagged flood condition reports verified by ML backend
Q18/Q19 Theme 4: Mismatch between official signals and ground reality	Institutional trust deficit	Ground-Truth Calibration Engine — ML model ingesting crowdsourced reports to adjust susceptibility ratings

Pillar 3: Sociotechnical Implementation Challenges & UX/UI Recommendations

Challenge 1: Varying Digital Literacy Across Barangays

The survey's geographic diversity (14 barangays, multiple socioeconomic contexts) implies significant variation in smartphone literacy, data plan availability, and comfort with app-based systems. Older residents and those in lower-income barangays may struggle with feature-dense interfaces.

Recommendations:

- Design FloodSense with a "Crisis Mode" interface — a single-screen, large-text, high-contrast view activated during active flood events showing only 3 things: current risk level, nearest evacuation center, and one emergency action.
- Conduct co-design workshops with residents from Soldiers 4, Talaba 1, and Molino 3 before finalizing UX flows, ensuring the interface reflects actual user mental models.
- Support SMS-based fallback functionality for users without smartphones or data access — critical for reaching the most vulnerable residents.

Challenge 2: Connectivity Constraints During Peak Flood Events

Internet connectivity is frequently degraded or unavailable during heavy rainfall events in Philippine urban areas. An app that requires constant internet connectivity will fail precisely when it is most needed.

Recommendations:

- Implement offline caching of the most recent barangay risk map, evacuation center locations, and protocol guides, refreshed every 15 minutes when connectivity is available.
- Use lightweight data protocols (compressed GeoJSON, not heavy map tiles) to minimize data consumption and maintain functionality on 2G/3G connections.
- Enable SMS push alert integration via LGU partnership so critical risk level changes are delivered even without app connectivity.

Challenge 3: Institutional Trust Deficit (Q18 Theme 4)

Residents who have experienced the PAGASA-vs-ground-reality credibility gap will approach any officially-endorsed app with skepticism.

Recommendations:

- Integrate the community crowdsourced reporting feature from day one so residents see that the app incorporates their real-world observations — not just government data.
- Display a model confidence indicator alongside risk levels to demonstrate data provenance and transparency.
- Launch with a 30-day pre-flood-season calibration period (barangay awareness campaigns, test alerts, community reporter recruitment) before the first operational rainy season.

Challenge 4: Fatalistic Non-Adoption (Q18 Gap 5)

The "Sanay na kami" (we're used to it) response signals that a subset of residents may not perceive sufficient personal benefit to download and learn a new app.

Recommendations:

- Deploy the app through barangay captain endorsement and BDRRMO co-branding, giving it institutional legitimacy that overcomes individual skepticism.
- Run pre-season community demonstrations in each barangay's public assembly, showing real-time AR overlays using existing flood scenario data.
- Offer household incentive programs for early adopters who complete the onboarding tutorial and submit their first community flood report.

Challenge 5: Barangay-Level Data Gaps for ML Model Training

FloodSense's ML flood susceptibility model requires historical flood data, elevation data (DEM), drainage network data, and rainfall records at the barangay level. Data quality across all 14+ barangays may be inconsistent.

Recommendations:

- Partner with PAGASA, PHIVOLCS, and the Bacoor City BDRRMO for official data access under a formal research MOU, establishing the app as an academic-government co-development project.
- Use Sentinel-1 SAR imagery (free, cloud-penetrating) for historical flood inundation mapping across barangays where ground truth data is absent.
- Implement a tiered confidence display system: barangays with sufficient training data display ML-generated risk levels; data-sparse barangays display a "Limited Data" flag, setting honest user expectations.